Description

Illustration





Figure caption

1. Fly ash is a waste product from burning coal and can be used to improve the properties of concrete. 2. High volume fly ash concrete bricks. Images used under license from Shutterstock.com

The material

Fly ash is a by-product from coal-fired electricity generating power plants. The molten ash is entrained in the flue gas and cools rapidly, into spherical, glassy particles. The fly ash is collected in electrostatic precipitators. Fly ash is a valuable additive that makes concrete stronger, more durable and easier to work. Energy is saved by reducing the fossil fuel consumption required to produce competing products such as cement, lime and crushed stone, and to mine products such as gypsum, limestone, sand, clay and gravel.
High volume fly ash concrete (HVFC) is closer to ancient concrete than any other type. The concrete used in ancient Roman buildings contained a very high content of pozzolanic volcanic ash. The very fine powder of the ash provides an excellent compacting material for low cement content concrete. It is clear, given the longevity of some Roman buildings, that the durability of such a material is very good - given the right mix ratios and under the right conditions. Note that the properties here regard unreinforced concrete, rather than the significantly improved properties that can be achieved by reinforcement with concrete.

Compositional summary

0.8:1:1:4.67:1.33:5 water:Portland Cement:Flyash:Course Sand:3/8" Gravel:3/4" Rock
or
0.92:1:1.32:6.41:9.4 water:Portland Cement:Flyash:Fine aggregate:Coarse aggregate
Mix constituents and proportions are varied to produce widely ranging properties.

General properties

Density		1.83e3	-	2.2e3	kg/m^3
Price	*	0.0302	-	0.128	GBP/kg

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✗
Services	✗

Mechanical properties

Young's modulus	* 30	-	41	GPa
Shear modulus	* 13	-	17	GPa
Bulk modulus	* 19	-	24	GPa
Bending modulus	* 18	-	20	GPa
Poisson's ratio	* 0.2	-	0.24	
Yield strength (elastic limit)	* 1.5	-	2.5	MPa
Tensile strength	* 1.5	-	2.5	MPa
Compressive strength	45	-	58	MPa
Bending strength	* 1.5	-	2.5	MPa
Elongation	* 0	-	0.01	% strain
Hardness - Vickers	* 12	-	16	HV
Fatigue strength at 10^7 cycles	* 0.9	-	1.4	MPa
Fracture toughness	* 0.35	-	0.45	MPa.m^0.5
Mechanical loss coefficient (tan delta)	* 0.01	-	0.03	

Thermal and combustion properties

Thermal conductor or insulator?	Poor insulator			
Thermal resistivity	0.8	-	0.9	m.°C/W
Thermal expansion coefficient	12	-	13	µstrain/°C
Specific heat capacity	* 813	-	867	J/kg.°C
Melting point	1.28e3	-	1.48e3	°C
Maximum service temperature	877	-	977	°C
Minimum service temperature	-160	-	-150	°C
Flammability	Non-flammable			
Emissivity	0.63	-	0.97	

Hygro-thermal properties

Water vapor permeability	2e-12	-	6.01e-12	kg.m/s.m^2.Pa
Frost resistance	Average			

Electrical properties

Electrical conductor or insulator?	Poor insulator

Electrical resistivity	* 2e11	-	2e12	µohm.cm
Dielectric constant (relative permittivity)	* 8	-	12	
Dissipation factor (dielectric loss tangent)	* 0.001	-	0.01	
Dielectric strength (dielectric breakdown)	* 1.4	-	1.6	MV/m

Optical properties

Natural color

Gray

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Poor
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Moldability	3	-	4
Machinability	1	-	2

Durability

Water (fresh)	Excellent
Water (salt)	Acceptable
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Limited use
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Industrial atmosphere	Limited use
Rural atmosphere	Excellent
Marine atmosphere	Acceptable

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.108	-	0.119	kg/kg
Embodied energy (virgin grade production)	* 0.9	-	0.995	MJ/kg
Water Usage	* 2.4	-	2.65	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 21.8	-	24.6	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO2-eq)	* 1.42	-	1.61	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.0108	-	0.0119	kg/kg
Embodied energy, recycling	* 0.126	-	0.139	MJ/kg
Recycle fraction in current supply	14			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Fly ash is commonly used as an Alternative Cementing Material (ACM) in concrete as a partial replacement for Portland cement (CEM I) - see Cement.. The percentage of fly ash commonly ranges from 15 to 35%.
High volume fly ash concrete - where the fly ash content is up to 55% (termed CEM IV/B-V in EN 197-1), can be used for special applications, offering significant benefit in terms of improved durability in sulfate and chloride ion exposures and very low heat of hydration, at the sacrifice of lower rate of strength gain.

The long-term strength and permeability are likely to be better than CEM I concrete of the same ultimate strength, provided it is thoroughly cured.

High volume fly ash (CEM IV/B-V) is called pozzolanic cement, not to be confused with Portland-pozzolana cement (e.g. CEM II/B-P) or Portland-fly-ash cement (e.g. CEM II/B-V) that have a maximum replacement level of 35%.

Technical notes

Fly ash, also known as flue ash and pulverized fuel ash, consists of ground ash and soot produced from burning coal. It is very fine, with a spherical particle shape and when incorporated into a concrete mix can be expected to provide the following benefits:
- Improved workability and pumpability
- Reduced water demand and bleeding
- Improved shrinkage resistance
- Improved resistance to sulfate and chloride attack
- Reduced potential for Alkali-Silica Reaction

Sulfate resistance is also determined by the composition of the fly ash and the highest resistance (e.g. CEM II/B-V+SR) has a lower alumina content, as defined in the UK in BS 8500-2

Note that the properties here regard unreinforced concrete, rather than the significantly improved properties that can be achieved by reinforcement with steel bar or fibres.

Typical uses

Paths, floors, foundations, road stabilization and large structures such as dams and pile-caps.

Related standards

The properties of Cement and Concrete vary a great deal based on composition, age and processing. Standards exist to help ensure that consistent material properties can be achieved. As temperature ranges, humidity, raw materials and building practices vary from country to country so do standards. Here you can see a list of the main US, EU and UK standards related to this material. Other countries have their own standards, for example: Japan www.jsa.or.jp/en, China www.gbstandards.org, Australia www.standards.org.au.

AS 3972: General purpose and blended cements
ASTM C150: Standard Specification for Portland Cement
ASTM C1697: Standard Specification for Blended Supplementary Cementitious Materials
ASTM C618: Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
EN 14227-3: Hydraulically bound mixtures. Specifications. Fly ash bound granular mixtures
EN 14227-4: Hydraulically bound mixtures. Specifications. Fly ash for hydraulically bound mixtures
EN 197-1: Cement. Composition, specifications and conformity criteria for common cements
EN 206: Concrete. Specification, performance, production and conformity
EN 450: Fly ash for concrete
BS 8500: Concrete. Complementary British Standard to BS EN 206

Links

Producers

Reference